Lesson 62: RKHS and the Representer Theorem

AI/ML Mathematics

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1 Motivation and Overview

Consider the problem of learning a function f from data $(x_1, y_1), \ldots, (x_n, y_n)$ by minimizing a regularized objective

$$\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \ell(y_i, f(x_i)) + \lambda\, \Omega(f),$$

where $\mathcal{F}$ is a function class, ℓ is a loss function, and $\Omega(f)$ is a regularizer penalizing function complexity. If $\mathcal{F}$ is a finite-dimensional space (e.g., polynomials of degree $\leq p$), the optimization reduces to a parameter search. But what if we want $\mathcal{F}$ to be a rich, infinite-dimensional space of functions—large enough to approximate any continuous function, yet structured enough that the optimization remains tractable?

Reproducing kernel Hilbert spaces (RKHS) provide exactly this. An RKHS $\mathcal{H}_k$ is an infinite-dimensional Hilbert space of functions equipped with a norm $\|f\|_{\mathcal{H}_k}$ that controls pointwise values: $|f(x)| \leq C_x \|f\|_{\mathcal{H}_k}$. The *representer theorem* shows that despite the infinite-dimensional search space, the minimizer of any regularized empirical risk over $\mathcal{H}_k$ lies in the n-dimensional subspace $\operatorname{span}\{k(x_1, \cdot), \ldots, k(x_n, \cdot)\}$. The tools developed in this lesson will let us reduce infinite-dimensional optimization to finite-dimensional linear algebra, providing the mathematical foundation for kernel ridge regression, support vector machines, and Gaussian process regression.

Why RKHS Theory Matters for RL/ML

- **Kernel ridge regression / GP regression:** The representer theorem reduces the infinite-dimensional minimization to solving an $n \times n$ linear system $(k + \lambda I)\alpha = y$. This is the computational backbone of Gaussian process regression with squared-error loss.
- **Support vector machines:** The SVM dual problem searches over $\alpha \in \mathbb{R}^n$ rather than $f \in \mathcal{H}_k$, a direct consequence of the representer theorem.
- **RKHS-based RL:** Kernel temporal difference learning and kernel Bellman equations represent value functions in an RKHS, with convergence guarantees inherited from RKHS norm control.
- **Neural tangent kernel theory:** In the infinite-width regime, the neural network function lies in the RKHS of the NTK, and training dynamics are governed by kernel ridge regression in this RKHS.

Prerequisites. Positive definite kernels, Mercer's theorem, and the kernel trick (Lesson 61). Hilbert spaces, orthogonal projections, and the Riesz representation theorem (Lesson 19, Lesson 20). Completeness and convergence in normed spaces (Lesson 16).

Learning objectives. After completing this lesson, the student will be able to:

- *Construct* the RKHS from a positive definite kernel and *prove* the reproducing property.
- *State* and *prove* Aronszajn's theorem on the uniqueness of the RKHS.
- *Derive* the norm-controls-pointwise-values bound and *characterize* RKHS elements via Mercer expansions.
- *State* and *prove* the representer theorem for regularized empirical risk minimization.
- *Apply* the representer theorem to derive the closed-form solution for kernel ridge regression.
- *Analyze* the excess risk of kernel ridge regression and *compute* the optimal regularization parameter.

2 References

- Bach [2024, Chapter 5] – RKHS construction, representer theorem, and kernel ridge regression in the learning theory framework.
- Aronszajn [1950] – Original paper establishing the abstract theory of reproducing kernels and RKHS.
- Steinwart and Christmann [2008, Chapters 4–5] – Rigorous treatment of RKHS theory, including interpolation and approximation properties.
- Schölkopf and Smola [2002, Chapter 2] – RKHS, representer theorem, and applications to SVMs.
- Mohri et al. [2018, Chapter 6] – Kernel methods and generalization bounds for RKHS-based algorithms.

3 RKHS Construction

We construct the reproducing kernel Hilbert space from a positive definite kernel in three steps: build a pre-Hilbert space, verify the reproducing property, and complete.

Definition 3.1 (Reproducing Kernel Hilbert Space)**.** Let $\mathcal{X}$ be a nonempty set. A *reproducing kernel Hilbert space* (RKHS) on $\mathcal{X}$ is a Hilbert space H of functions $f\colon \mathcal{X} \to \mathbb{R}$ such that for every $x \in \mathcal{X}$, the *evaluation functional* $\delta_x\colon H \to \mathbb{R}$ defined by $\delta_x(f) = f(x)$ is bounded (equivalently, continuous). That is, for each x there exists $C_x > 0$ such that

$$|f(x)| \leq C_x \|f\|_H \quad \text{for all } f \in H. \tag{1}$$

Remark 3.2. Not every Hilbert space of functions is an RKHS. For example, $L^2([0,1])$ is a Hilbert space of (equivalence classes of) functions, but point evaluation is not continuous: one can modify a function on a set of measure zero without changing its L^2 norm, so $f(x)$ is not well-defined as a functional on L^2. The RKHS condition ensures that functions in H have well-defined pointwise values.

Definition 3.3 (Reproducing Kernel)**.** A function $k\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is a *reproducing kernel* of a Hilbert space H of functions on $\mathcal{X}$ if:

(RK1) For each $x \in \mathcal{X}$, the function $k_x := k(x, \cdot)$ belongs to H.

(RK2) **Reproducing property:** For each $x \in \mathcal{X}$ and each $f \in H$:

$$f(x) = \langle f, k_x \rangle_H. \tag{2}$$

Remark 3.4. Applying the reproducing property twice: $k(x, x') = k_{x'}(x) = \langle k_{x'}, k_x \rangle_H = \langle k_x, k_{x'} \rangle_H$. This shows that the reproducing kernel is the inner product between the "canonical feature vectors" k_x and $k_{x'}$ in H.

We now construct the RKHS from a given pd kernel.

Theorem 3.5 (RKHS Construction)**.** *Let $k\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a positive definite kernel. There exists a unique Hilbert space $\mathcal{H}_k$ of functions on $\mathcal{X}$ for which k is the reproducing kernel.*

Proof. **Step 1: Define the pre-Hilbert space.** Let H_0 be the set of all finite linear combinations of kernel sections:

$$H_0 = \left\{ f = \sum_{i=1}^{m} \alpha_i k_{x_i} \;\middle|\; m \in \mathbb{N},\ \alpha_i \in \mathbb{R},\ x_i \in \mathcal{X} \right\},$$

where $k_{x_i}(\cdot) = k(x_i, \cdot)$. Each element of H_0 is a function $f\colon \mathcal{X} \to \mathbb{R}$.

Step 2: Define the inner product. For $f = \sum_{i=1}^{m} \alpha_i k_{x_i}$ and $g = \sum_{j=1}^{l} \beta_j k_{x'_j}$, define

$$\langle f, g \rangle_{H_0} = \sum_{i=1}^{m} \sum_{j=1}^{l} \alpha_i \beta_j \, k(x_i, x'_j). \tag{3}$$

We must verify this is well-defined (independent of the representation). Suppose $f = \sum_{i=1}^{m} \alpha_i k_{x_i} = 0$ as a function. Then for any $g = \sum_{j=1}^{l} \beta_j k_{x'_j}$:

$$\langle f, g \rangle_{H_0} = \sum_{i=1}^{m} \alpha_i \sum_{j=1}^{l} \beta_j k(x_i, x'_j) = \sum_{i=1}^{m} \alpha_i g(x_i) = 0,$$

since $\sum_{j=1}^{l} \beta_j k(x_i, x'_j) = g(x_i)$ and if $f = 0$ as a function then in particular $\sum_i \alpha_i g(x_i)$ can be shown to vanish by taking $g = f$ to get $\langle f, f \rangle_{H_0} = \sum_i \alpha_i f(x_i) = 0$ (since $f = 0$). By bilinearity, the inner product is independent of representation.

Step 3: Verify positivity. For $f = \sum_{i=1}^{m} \alpha_i k_{x_i}$:

$$\langle f, f \rangle_{H_0} = \sum_{i=1}^{m} \sum_{j=1}^{m} \alpha_i \alpha_j k(x_i, x_j) \geq 0,$$

by positive definiteness of k. If $\langle f, f \rangle_{H_0} = 0$, then for any $x \in \mathcal{X}$:

$$|f(x)|^2 = \left| \sum_{i=1}^{m} \alpha_i k(x_i, x) \right|^2 = |\langle f, k_x \rangle_{H_0}|^2 \leq \langle f, f \rangle_{H_0} \langle k_x, k_x \rangle_{H_0} = 0,$$

by Cauchy–Schwarz (which holds in any semi-inner product space). Hence $f = 0$ as a function.

Step 4: Verify the reproducing property in H_0. For $f = \sum_{i=1}^{m} \alpha_i k_{x_i}$ and any $x \in \mathcal{X}$:

$$\langle f, k_x \rangle_{H_0} = \sum_{i=1}^{m} \alpha_i k(x_i, x) = f(x).$$

Step 5: Complete H_0 to a Hilbert space $\mathcal{H}_k$. The space $(H_0, \langle \cdot, \cdot \rangle_{H_0})$ is a pre-Hilbert space. Let $\mathcal{H}_k$ be its completion: every element of $\mathcal{H}_k$ is the limit of a Cauchy sequence (f_n) in H_0. We must verify that elements of $\mathcal{H}_k$ are still functions on $\mathcal{X}$ and the reproducing property extends.

For a Cauchy sequence (f_n) in H_0, the reproducing property gives $|f_n(x) - f_m(x)| = |\langle f_n - f_m, k_x \rangle_{H_0}| \leq \|f_n - f_m\|_{H_0} \sqrt{k(x,x)}$. Hence $(f_n(x))$ is Cauchy in $\mathbb{R}$ for each x, so $f(x) := \lim_n f_n(x)$ exists. This defines the limit as a function $f\colon \mathcal{X} \to \mathbb{R}$. The reproducing property $f(x) = \langle f, k_x \rangle_{\mathcal{H}_k}$ follows by continuity of the inner product.

Step 6: Uniqueness. Suppose H' is another Hilbert space of functions on $\mathcal{X}$ with k as its reproducing kernel. Then $H_0 \subseteq H'$ (since $k_x \in H'$ for all x), and the inner products agree on H_0 (by the reproducing property: $\langle k_x, k_{x'} \rangle_{H'} = k(x, x') = \langle k_x, k_{x'} \rangle_{H_0}$). Since H_0 is dense in $\mathcal{H}_k$, and H' is complete, we have $\mathcal{H}_k \subseteq H'$. By a symmetric argument using the density of H_0 in H', we get $H' \subseteq \mathcal{H}_k$. Hence $\mathcal{H}_k = H'$ with equal inner products. $\square$

Aronszajn's Theorem

There is a one-to-one correspondence between positive definite kernels and RKHS: every pd kernel k determines a unique RKHS $\mathcal{H}_k$, and conversely, every RKHS H has a unique reproducing kernel (given by $k(x, x') = \langle k_x, k_{x'} \rangle_H$ where k_x is the Riesz representer of δ_x). This is *Aronszajn's theorem* [Aronszajn, 1950].

Example 3.6 (RKHS of the Linear Kernel)**.** For $k(x, x') = x^\top x'$ on $\mathbb{R}^d$, the RKHS is $\mathcal{H}_k = \{f(x) = w^\top x : w \in \mathbb{R}^d\}$ with inner product $\langle f_w, f_v \rangle_{\mathcal{H}_k} = w^\top v$. The reproducing property: $f_w(x) = w^\top x = \langle f_w, f_x \rangle_{\mathcal{H}_k}$ (here $k_x(\cdot) = x^\top(\cdot) = f_x$). This is just $\mathbb{R}^d$ relabeled as a space of linear functions.

Example 3.7 (RKHS of the Polynomial Kernel)**.** For the degree-p polynomial kernel $k(x, x') = (x^\top x')^p$ on $\mathbb{R}^d$, the RKHS consists of all homogeneous polynomials of degree p, equipped with a specific inner product. For $d = 2$ and $p = 2$: $k(x, x') = (x_1 x_1' + x_2 x_2')^2 = x_1^2 (x_1')^2 + 2x_1 x_2 x_1' x_2' + x_2^2 (x_2')^2$, and $\mathcal{H}_k = \{f(x) = ax_1^2 + bx_1 x_2 + cx_2^2 : a, b, c \in \mathbb{R}\}$ with $\|f\|_{\mathcal{H}_k}^2 = a^2 + b^2/2 + c^2$. Note that the cross-term $x_1 x_2$ has half the norm penalty of the pure terms, reflecting the combinatorial coefficient in the kernel expansion.

The RKHS construction provides a concrete Hilbert space in which kernel methods operate. We next study the key structural properties of this space.

4 Properties of RKHS

4.1 Norm Controls Pointwise Values

The fundamental property of an RKHS is that the norm provides uniform control over pointwise function values.

Theorem 4.1 (Pointwise Bound)**.** *For any $f \in \mathcal{H}_k$ and any $x \in \mathcal{X}$:*

$$|f(x)| \leq \|f\|_{\mathcal{H}_k} \sqrt{k(x, x)}. \tag{4}$$

Proof. By the reproducing property and the Cauchy–Schwarz inequality:

$$|f(x)| = |\langle f, k_x \rangle_{\mathcal{H}_k}| \leq \|f\|_{\mathcal{H}_k} \|k_x\|_{\mathcal{H}_k} = \|f\|_{\mathcal{H}_k} \sqrt{\langle k_x, k_x \rangle_{\mathcal{H}_k}} = \|f\|_{\mathcal{H}_k} \sqrt{k(x, x)}.$$

□

Remark 4.2. The bound is tight: equality holds for $f = k_x / \|k_x\|_{\mathcal{H}_k}$ (assuming $k(x, x) > 0$). This function is the "peakiest" unit-norm function at x.

Corollary 4.3 (Uniform Pointwise Bound on Compact Domains)**.** *If $\mathcal{X}$ is compact and k is continuous, then $\sup_{x \in \mathcal{X}} |f(x)| \leq M \|f\|_{\mathcal{H}_k}$ where $M = \sup_{x \in \mathcal{X}} \sqrt{k(x, x)} < \infty$. In particular, convergence in RKHS norm implies uniform convergence.*

Proof. Since k is continuous on the compact set $\{(x, x) : x \in \mathcal{X}\}$, $M := \sup_x \sqrt{k(x, x)} < \infty$. Apply Theorem 4.1. □

Norm Control in Function Approximation

In value function approximation, we seek $\hat{V} \in \mathcal{H}_k$ approximating the true value function V^*. The pointwise bound ensures that if $\|\hat{V} - V^*\|_{\mathcal{H}_k}$ is small, then $|\hat{V}(s) - V^*(s)|$ is small *simultaneously at all states s*. This is much stronger than L^2 convergence, which only

controls average error. The norm penalty $\lambda\|f\|_{\mathcal{H}_k}^2$ in regularized methods thus directly controls the pointwise approximation quality.

4.2 Convergence and Inclusion

Proposition 4.4 (RKHS Convergence Implies Pointwise Convergence)**.** *If $f_n \to f$ in $\mathcal{H}_k$, then $f_n(x) \to f(x)$ for every $x \in \mathcal{X}$. If additionally $\mathcal{X}$ is compact and k is continuous, the convergence is uniform.*

Proof. $|f_n(x) - f(x)| \leq \|f_n - f\|_{\mathcal{H}_k}\sqrt{k(x,x)} \to 0$ by Theorem 4.1. Uniformity follows from Corollary 4.3. □

Theorem 4.5 (RKHS Inclusion)**.** *Let k_1, k_2 be pd kernels on $\mathcal{X}$ with $k_2 - k_1$ positive definite. Then $\mathcal{H}_{k_1} \subseteq \mathcal{H}_{k_2}$ and $\|f\|_{\mathcal{H}_{k_2}} \leq \|f\|_{\mathcal{H}_{k_1}}$ for all $f \in \mathcal{H}_{k_1}$.*

Proof. Write $k_2 = k_1 + k_3$ where $k_3 = k_2 - k_1$ is pd. The RKHS of k_2 satisfies $\mathcal{H}_{k_2} = \mathcal{H}_{k_1} + \mathcal{H}_{k_3}$ (as a sum of Hilbert spaces). For any $f \in \mathcal{H}_{k_1}$, write $f = f + 0$ with $f \in \mathcal{H}_{k_1}$ and $0 \in \mathcal{H}_{k_3}$. Then

$$\|f\|_{\mathcal{H}_{k_2}}^2 = \inf_{f=g+h}\left(\|g\|_{\mathcal{H}_{k_1}}^2 + \|h\|_{\mathcal{H}_{k_3}}^2\right) \leq \|f\|_{\mathcal{H}_{k_1}}^2 + 0.$$ □

4.3 Mercer Characterization of RKHS Elements

When Mercer's theorem applies (continuous kernel on a compact domain with a full-support measure μ), we can characterize RKHS elements through their Mercer expansion.

Theorem 4.6 (Mercer Characterization)**.** *Under the setting of Mercer's theorem (Lesson 61, Theorem 5.4), let $k(x,x') = \sum_{j=1}^{\infty} \lambda_j \phi_j(x)\phi_j(x')$ be the Mercer expansion. Then $f \in \mathcal{H}_k$ if and only if $f = \sum_{j=1}^{\infty} a_j\phi_j$ (convergence in $L^2(\mu)$ and pointwise) with*

$$\|f\|_{\mathcal{H}_k}^2 = \sum_{j=1}^{\infty} \frac{a_j^2}{\lambda_j} < \infty, \tag{5}$$

where the sum runs only over j with $\lambda_j > 0$.

Proof. **("If" direction.)** Suppose $f = \sum_{j=1}^{\infty} a_j\phi_j$ with $\sum_j a_j^2/\lambda_j < \infty$. Define the partial sums $f_N = \sum_{j=1}^{N} a_j\phi_j$. We can write f_N as a finite combination of kernel sections: since $\phi_j = \lambda_j^{-1}T_k\phi_j$, we have $\phi_j(x) = \lambda_j^{-1}\int k(x,x')\phi_j(x')\,d\mu(x')$, so ϕ_j is in the L^2-closure of kernel sections. More directly, the reproducing property gives $\langle k_x, \phi_j\rangle_{\mathcal{H}_k} = \phi_j(x)$, and one can verify that $\langle \phi_i, \phi_j\rangle_{\mathcal{H}_k} = \delta_{ij}/\lambda_j$ (the Mercer eigenfunctions, when normalized in L^2, are orthogonal in $\mathcal{H}_k$ with squared norms $1/\lambda_j$).

To see this, compute: $\langle \phi_i, \phi_j\rangle_{\mathcal{H}_k} = \phi_i(x)|_x = \ldots$ More carefully: by the reproducing property, $\phi_j(x) = \langle \phi_j, k_x\rangle_{\mathcal{H}_k}$. Using the Mercer expansion of k_x: $k_x = \sum_l \lambda_l\phi_l(x)\phi_l$, so $\langle \phi_j, k_x\rangle_{\mathcal{H}_k} = \sum_l \lambda_l\phi_l(x)\langle \phi_j, \phi_l\rangle_{\mathcal{H}_k}$. For this to equal $\phi_j(x)$ for all x, we need $\lambda_l\langle \phi_j, \phi_l\rangle_{\mathcal{H}_k} = \delta_{jl}$, giving $\langle \phi_j, \phi_l\rangle_{\mathcal{H}_k} = \delta_{jl}/\lambda_j$.

Therefore $\|f_N\|_{\mathcal{H}_k}^2 = \sum_{j=1}^{N} a_j^2/\lambda_j$. The sequence (f_N) is Cauchy in $\mathcal{H}_k$ since $\|f_N - f_M\|_{\mathcal{H}_k}^2 = \sum_{j=M+1}^{N} a_j^2/\lambda_j \to 0$ as $M, N \to \infty$ (tail of a convergent series). By completeness of $\mathcal{H}_k$, $f_N \to f^*$ in $\mathcal{H}_k$ for some $f^* \in \mathcal{H}_k$. By Proposition 4.4, $f^*(x) = \lim_N f_N(x) = \sum_j a_j\phi_j(x) = f(x)$. Hence $f = f^* \in \mathcal{H}_k$ and $\|f\|_{\mathcal{H}_k}^2 = \sum_j a_j^2/\lambda_j$.

("Only if" direction.) Let $f \in \mathcal{H}_k$. Since $\{\phi_j/\sqrt{\lambda_j}\}$ is an orthonormal system in $\mathcal{H}_k$, we have $f = \sum_j \langle f, \phi_j/\sqrt{\lambda_j}\rangle_{\mathcal{H}_k} \cdot \phi_j/\sqrt{\lambda_j} + f^\perp$, where $f^\perp$ is orthogonal to all ϕ_j. But $f^\perp(x) = \langle f^\perp, k_x\rangle_{\mathcal{H}_k} = \sum_l \lambda_l\phi_l(x)\langle f^\perp, \phi_l\rangle_{\mathcal{H}_k} = 0$ for all x (since $\langle f^\perp, \phi_l\rangle_{\mathcal{H}_k} = 0$). So $f^\perp = 0$. Setting $a_j = \langle f, \phi_j\rangle_{\mathcal{H}_k} \cdot \sqrt{\lambda_j}$, we get $f = \sum_j a_j\phi_j$ and $\|f\|_{\mathcal{H}_k}^2 = \sum_j a_j^2/\lambda_j$. □

Example 4.7 (RKHS of the Gaussian Kernel)**.** The Gaussian kernel on a compact domain has eigenvalues λ_j decaying exponentially fast (as noted in Lesson 61, Example 5.5). The Mercer characterization shows that $f \in \mathcal{H}_k$ requires $\sum_j a_j^2/\lambda_j < \infty$, meaning the Fourier-like coefficients a_j must decay *faster* than $\sqrt{\lambda_j}$, i.e., faster than exponentially. This imposes extreme smoothness: the RKHS of the Gaussian kernel consists of C^∞ functions (in fact, real-analytic functions).

Conversely, this means the Gaussian RKHS is a relatively "small" function space. Functions with discontinuities, corners, or even finite-order differentiability do not lie in the Gaussian RKHS. This explains why Gaussian kernel methods may underperform on non-smooth target functions.

RKHS Norm vs. L^2 Norm

The RKHS norm $\|f\|_{\mathcal{H}_k}^2 = \sum_j a_j^2/\lambda_j$ penalizes high-frequency components (large j, small λ_j) much more heavily than the L^2 norm $\|f\|_{L^2}^2 = \sum_j a_j^2$. A function can have small L^2 norm but infinite RKHS norm if its high-frequency content decays too slowly. Regularizing with $\|f\|_{\mathcal{H}_k}^2$ thus imposes a smoothness bias.

We now have a complete characterization of the RKHS. The central computational result of kernel methods is the representer theorem, which we develop next.

5 The Representer Theorem

The representer theorem is the computational cornerstone of kernel methods. It shows that the solution to any regularized empirical risk minimization problem over an RKHS is a finite linear combination of kernel sections centered at the training points, regardless of the (possibly infinite) dimensionality of the RKHS.

Theorem 5.1 (Representer Theorem)**.** *Let $\mathcal{H}_k$ be the RKHS associated with a pd kernel k on $\mathcal{X}$. Let $x_1, \ldots, x_n \in \mathcal{X}$ be training points and consider the optimization problem*

$$\min_{f \in \mathcal{H}_k} J(f) := c\big(f(x_1), \ldots, f(x_n)\big) + g\big(\|f\|_{\mathcal{H}_k}\big), \tag{6}$$

where $c\colon \mathbb{R}^n \to \mathbb{R}$ is an arbitrary function (the empirical cost) and $g\colon [0,\infty) \to \mathbb{R}$ is strictly increasing (the regularizer). Then if (6) has a minimizer, it has one of the form

$$f^*(x) = \sum_{i=1}^n \alpha_i k(x_i, x) \tag{7}$$

for some $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$.

Proof. The proof uses orthogonal decomposition in the Hilbert space $\mathcal{H}_k$.

Step 1: Decompose. Let $S = \operatorname{span}\{k_{x_1}, \ldots, k_{x_n}\}$ be the (closed, finite-dimensional) subspace of $\mathcal{H}_k$ spanned by the kernel sections at the training points. Any $f \in \mathcal{H}_k$ can be decomposed as

$$f = f_S + f_\perp, \quad f_S \in S,\ f_\perp \in S^\perp,$$

where $S^\perp = \{h \in \mathcal{H}_k : \langle h, g\rangle_{\mathcal{H}_k} = 0 \text{ for all } g \in S\}$.

Step 2: Show $f_\perp$ does not affect evaluations. For any x_i ($i = 1, \ldots, n$):

$$f_\perp(x_i) = \langle f_\perp, k_{x_i}\rangle_{\mathcal{H}_k} = 0,$$

since $f_\perp \in S^\perp$ and $k_{x_i} \in S$. Therefore $f(x_i) = f_S(x_i) + f_\perp(x_i) = f_S(x_i)$ for all i.

Step 3: Show $f_\perp$ only increases the norm. By the Pythagorean theorem:

$$\|f\|_{\mathcal{H}_k}^2 = \|f_S\|_{\mathcal{H}_k}^2 + \|f_\perp\|_{\mathcal{H}_k}^2 \geq \|f_S\|_{\mathcal{H}_k}^2,$$

with equality iff $f_\perp = 0$.

Step 4: Conclude. Combining Steps 2 and 3:

$$J(f) = c(f(x_1), \ldots, f(x_n)) + g(\|f\|_{\mathcal{H}_k}) = c(f_S(x_1), \ldots, f_S(x_n)) + g(\|f\|_{\mathcal{H}_k}).$$

Since g is strictly increasing and $\|f\|_{\mathcal{H}_k} \geq \|f_S\|_{\mathcal{H}_k}$:

$$J(f) \geq c(f_S(x_1), \ldots, f_S(x_n)) + g(\|f_S\|_{\mathcal{H}_k}) = J(f_S).$$

Strict inequality holds whenever $f_\perp \neq 0$. Hence any minimizer must satisfy $f_\perp = 0$, i.e., $f^* = f_S^* \in S$. Since $S = \text{span}\{k_{x_1}, \ldots, k_{x_n}\}$, we have $f^*(x) = \sum_{i=1}^n \alpha_i k(x_i, x)$. □

Representer Theorem: From Infinite to Finite Dimensions

The representer theorem reduces the infinite-dimensional optimization $\min_{f \in \mathcal{H}_k} J(f)$ to the finite-dimensional problem $\min_{\alpha \in \mathbb{R}^n} J_\alpha(\alpha)$, where

$$J_\alpha(\alpha) = c\left(\sum_i \alpha_i k(x_i, x_1), \ldots, \sum_i \alpha_i k(x_i, x_n)\right) + g\left(\sqrt{\alpha^\top K \alpha}\right),$$

with $K_{ij} = k(x_i, x_j)$. The search space dimension equals the number of training points n, regardless of the RKHS dimension (which may be infinite).

Example 5.2 (Representer Theorem for Squared Loss)**.** For the squared loss $c(z_1, \ldots, z_n) = \frac{1}{n}\sum_{i=1}^n (y_i - z_i)^2$ and Tikhonov regularization $g(t) = \lambda t^2$, the representer theorem gives

$$f^*(x) = \sum_{i=1}^n \alpha_i k(x_i, x), \qquad \min_{\alpha \in \mathbb{R}^n} \frac{1}{n}\|y - K\alpha\|^2 + \lambda\, \alpha^\top K \alpha.$$

This is kernel ridge regression, developed in detail in Section 6.

Example 5.3 (Representer Theorem for SVM)**.** For the support vector machine with hinge loss $c(z_1, \ldots, z_n) = \frac{1}{n}\sum_{i=1}^n \max(0, 1 - y_i z_i)$ (with $y_i \in \{-1, +1\}$), the representer theorem gives $f^* = \sum_i \alpha_i k(x_i, \cdot)$. The optimal α is found by solving a quadratic program. The sparsity of the SVM solution (most $\alpha_i = 0$) is an additional property of the hinge loss, not of the representer theorem itself.

Representer Theorem and Nonparametric RL

In kernel-based reinforcement learning, the value function is represented as $\hat{V}(s) = \sum_{i=1}^n \alpha_i k(s_i, s)$ where $s_1, \ldots, s_n$ are visited states. The representer theorem guarantees that this finite representation is optimal among all functions in the (infinite-dimensional) RKHS, provided the objective has the form of a regularized empirical cost evaluated at the visited states. This justifies the kernel Bellman equation approach: $\hat{V} = \arg\min_{f \in \mathcal{H}_k} \sum_i (f(s_i) - r_i - \gamma f(s_i'))^2 + \lambda \|f\|_{\mathcal{H}_k}^2$.

6 Kernel Ridge Regression

We apply the representer theorem to derive the complete solution for kernel ridge regression, then analyze its statistical properties.

6.1 Closed-Form Solution

Definition 6.1 (Kernel Ridge Regression)**.** Given training data $(x_1, y_1), \ldots, (x_n, y_n) \in \mathcal{X} \times \mathbb{R}$ and regularization parameter $\lambda > 0$, *kernel ridge regression* (KRR) solves

$$f^* = \arg\min_{f \in \mathcal{H}_k} \frac{1}{n} \sum_{i=1}^{n} (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}_k}^2. \tag{8}$$

Theorem 6.2 (KRR Solution)**.** *The minimizer of* (8) *is* $f^*(x) = \sum_{j=1}^{n} \alpha_j^* k(x_j, x)$ *where*

$$\alpha^* = (K + n\lambda I)^{-1} y, \tag{9}$$

with $K_{ij} = k(x_i, x_j)$ *and* $y = (y_1, \ldots, y_n)^\top$*. The matrix* $K + n\lambda I$ *is positive definite (hence invertible) for any* $\lambda > 0$*.*

Proof. By the representer theorem (Theorem 5.1), the minimizer has the form $f^*(x) = \sum_j \alpha_j k(x_j, x)$. Substituting into (8): the prediction at x_i is $f(x_i) = \sum_j \alpha_j k(x_j, x_i) = (K\alpha)_i$, and the RKHS norm is

$$\|f\|_{\mathcal{H}_k}^2 = \left\langle \sum_i \alpha_i k_{x_i}, \sum_j \alpha_j k_{x_j} \right\rangle_{\mathcal{H}_k} = \sum_{i,j} \alpha_i \alpha_j k(x_i, x_j) = \alpha^\top K \alpha.$$

The objective becomes:

$$J(\alpha) = \frac{1}{n} \|y - K\alpha\|^2 + \lambda \alpha^\top K \alpha = \frac{1}{n}(y^\top y - 2y^\top K\alpha + \alpha^\top K^2 \alpha) + \lambda \alpha^\top K \alpha.$$

Taking the gradient and setting it to zero:

$$\nabla_\alpha J = \frac{1}{n}(-2Ky + 2K^2\alpha) + 2\lambda K\alpha = \frac{2K}{n}(-y + K\alpha + n\lambda\alpha) = 0.$$

Since K may be singular, we need the factor $(-y + K\alpha + n\lambda\alpha) = 0$, giving $(K + n\lambda I)\alpha = y$. Since $K \succeq 0$ and $n\lambda > 0$, the matrix $K + n\lambda I$ is positive definite, hence invertible: $\alpha^* = (K + n\lambda I)^{-1} y$. □

Example 6.3 (KRR with the Gaussian Kernel)**.** Let $\mathcal{X} = \mathbb{R}$, $k(x, x') = \exp(-(x - x')^2/(2\sigma^2))$, and suppose we have $n = 5$ training points. The prediction at a test point x is

$$f^*(x) = \sum_{j=1}^{5} \alpha_j^* \exp\left(-\frac{(x_j - x)^2}{2\sigma^2}\right),$$

a weighted sum of Gaussian bumps centered at the training points. The weights α^* are determined by the linear system $(K + n\lambda I)\alpha^* = y$.

For $\sigma \to 0$ (narrow bumps), $K \to I$ and $\alpha^* \to y/(1 + n\lambda)$: the predictor interpolates (approximately) at the training points. For $\sigma \to \infty$ (wide bumps), $K \to \mathbf{1}\mathbf{1}^\top$ and the predictor approaches a constant function.

6.2 Statistical Analysis

We now analyze the excess risk of kernel ridge regression. The excess risk measures how much worse our estimator f^* performs compared to the best function in the RKHS.

Assumption 6.4 (Statistical Setting)**.** The data (x_i, y_i) are drawn i.i.d. from a distribution $\mathbb{P}$ on $\mathcal{X} \times \mathbb{R}$. The regression function is $f_0(x) = \mathbb{E}[Y \mid X = x]$. We assume $f_0 \in \mathcal{H}_k$, $\|f_0\|_{\mathcal{H}_k} \le B$, $|Y| \le M$ almost surely, and $k(x, x) \le \kappa^2$ for all $x \in \mathcal{X}$.

Definition 6.5 (Excess Risk)**.** The *excess risk* of an estimator $\hat{f}$ is

$$\mathcal{E}(\hat{f}) = \mathbb{E}_X[(\hat{f}(X) - f_0(X))^2] = \|\hat{f} - f_0\|^2_{L^2(\mathbb{P}_X)}, \tag{10}$$

where $\mathbb{P}_X$ is the marginal distribution of X.

Theorem 6.6 (Excess Risk Bound for KRR)**.** *Under Assumption 6.4, the KRR estimator $\hat{f}_\lambda$ with regularization parameter $\lambda > 0$ satisfies*

$$\mathbb{E}[\mathcal{E}(\hat{f}_\lambda)] \le C_1 \lambda B^2 + \frac{C_2 \kappa^2 M^2}{n\lambda}, \tag{11}$$

where $C_1, C_2 > 0$ are universal constants.

Proof. We decompose the excess risk into approximation error (bias) and estimation error (variance).

Step 1: Bias-variance decomposition. Define the "population" regularized risk minimizer $f_\lambda = \arg\min_{f \in \mathcal{H}_k} \mathbb{E}[(Y - f(X))^2] + \lambda \|f\|^2_{\mathcal{H}_k}$. Then

$$\mathcal{E}(\hat{f}_\lambda) \le 2\|f_\lambda - f_0\|^2_{L^2} + 2\|\hat{f}_\lambda - f_\lambda\|^2_{L^2}.$$

Step 2: Approximation error (bias). The population minimizer f_λ satisfies the optimality condition $(T + \lambda I) f_\lambda = T f_0$ where T is the integral operator of k with respect to $\mathbb{P}_X$. Then $f_\lambda - f_0 = -\lambda (T + \lambda I)^{-1} f_0$. Hence

$$\|f_\lambda - f_0\|^2_{L^2} = \lambda^2 \|(T + \lambda I)^{-1} f_0\|^2_{L^2} \le \lambda^2 \|f_0\|^2_{\mathcal{H}_k} / \lambda = \lambda B^2,$$

where we used $\|(T + \lambda I)^{-1}\|_{\mathcal{H}_k \to L^2} \le 1/\sqrt{\lambda}$ (since the eigenvalues of $(T + \lambda I)^{-1}$ in the RKHS-to-L^2 mapping are bounded by $\sqrt{\lambda_j}/(\lambda_j + \lambda) \le 1/(2\sqrt{\lambda})$, using $\sqrt{t}/(t + \lambda) \le 1/(2\sqrt{\lambda})$ for $t \ge 0$).

Step 3: Estimation error (variance). The estimation error $\|\hat{f}_\lambda - f_\lambda\|^2_{L^2}$ depends on the random fluctuation of the empirical risk around the population risk. Standard arguments (see Bach [2024, Chapter 5]) using the RKHS norm bound on the function class and concentration inequalities yield

$$\mathbb{E}[\|\hat{f}_\lambda - f_\lambda\|^2_{L^2}] \le \frac{C_2 \kappa^2 M^2}{n\lambda}.$$

The key ingredient is that the effective dimension of the search space (the number of eigenvalues of T above λ) controls the variance, and the RKHS norm bound controls the sup-norm of functions in the class via Theorem 4.1.

Step 4: Combine. Adding the two terms gives (11). □

Corollary 6.7 (Optimal Regularization)**.** *Minimizing the bound (11) over λ: the derivative with respect to λ is $C_1 B^2 - C_2 \kappa^2 M^2 / (n\lambda^2)$. Setting to zero gives the optimal*

$$\lambda^* = \left(\frac{C_2 \kappa^2 M^2}{C_1 B^2 n} \right)^{1/2} = O(n^{-1/2}), \tag{12}$$

yielding the excess risk bound

$$\mathbb{E}[\mathcal{E}(\hat{f}_{\lambda^*})] \le 2\sqrt{C_1 C_2} \cdot \frac{\kappa B M}{\sqrt{n}} = O(n^{-1/2}). \tag{13}$$

Proof. Direct substitution: with $\lambda^* = \kappa M \sqrt{C_2/(C_1 B^2 n)}$,

$$C_1 \lambda^* B^2 = C_1 B^2 \cdot \kappa M \sqrt{\frac{C_2}{C_1 B^2 n}} = \kappa B M \sqrt{\frac{C_1 C_2}{n}},$$

$$\frac{C_2 \kappa^2 M^2}{n\lambda^*} = \frac{C_2 \kappa^2 M^2}{n} \cdot \frac{\sqrt{C_1 B^2 n}}{\kappa M \sqrt{C_2}} = \kappa B M \sqrt{\frac{C_1 C_2}{n}}.$$

The two terms are equal, and the total bound is $2\kappa B M \sqrt{C_1 C_2 / n}$. □

KRR Rate with Eigenvalue Decay

The $O(n^{-1/2})$ rate is a worst-case bound. When the eigenvalues of the kernel integral operator T satisfy $\lambda_j = O(j^{-\beta})$ for some $\beta > 1$ (polynomial decay), the excess risk of KRR with optimal λ improves to $O(n^{-\beta/(\beta+1)})$. For the Gaussian kernel ($\beta = \infty$, exponential decay), the rate approaches $O(n^{-1} \log n)$, nearly parametric. This *eigenvalue-dependent rate* is a central result connecting the smoothness of the kernel to the statistical efficiency of kernel methods.

Example 6.8 (KRR with Sobolev Kernel)**.** The first-order Sobolev kernel on $[0, 1]$ is $k(x, x') = 1 + \min(x, x')$, whose RKHS is the Sobolev space $W^{1,2}([0, 1])$ of functions with square-integrable first derivatives. The eigenvalues decay as $\lambda_j = O(j^{-2})$ ($\beta = 2$). By the eigenvalue-dependent rate, KRR achieves excess risk $O(n^{-2/3})$, better than the generic $O(n^{-1/2})$ but slower than the Gaussian kernel rate. This reflects the fact that the Sobolev RKHS contains rougher functions than the Gaussian RKHS.

KRR and Gaussian Process Regression

Kernel ridge regression and Gaussian process (GP) regression with squared-error loss produce *identical predictions*. If we place a GP prior $f \sim \mathcal{GP}(0, k)$ and assume Gaussian noise $y_i = f(x_i) + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, the posterior mean is $\hat{f}(x) = k_x^\top (K + \sigma^2 I)^{-1} y$, which matches the KRR solution with $\lambda = \sigma^2/n$. The Bayesian interpretation provides uncertainty quantification (posterior variance) that KRR alone does not.

7 Exercises

Exercise 7.1 (Reproducing Property Verification)**.** ($*$)
Let $k(x, x') = 1 + xx'$ on $\mathcal{X} = \mathbb{R}$.

(a) Describe the RKHS $\mathcal{H}_k$ as a set of functions and specify the inner product.

(b) Verify the reproducing property directly: for $f(x) = a + bx \in \mathcal{H}_k$, compute $\langle f, k_t \rangle_{\mathcal{H}_k}$ and confirm it equals $f(t)$.

Exercise 7.2 (RKHS Norm Computation)**.** ($*$)
For the kernel $k(x, x') = (x^\top x')^2$ on $\mathbb{R}^2$, compute $\|f\|_{\mathcal{H}_k}$ for the function $f(x) = x_1^2 - x_2^2$. Express your answer in terms of the Mercer expansion of k.

Exercise 7.3 (Representer Theorem Application)**.** ($**$)
Consider the regularized absolute-deviation problem

$$\min_{f \in \mathcal{H}_k} \frac{1}{n} \sum_{i=1}^{n} |y_i - f(x_i)| + \lambda \|f\|_{\mathcal{H}_k}^2.$$

(a) Verify that the representer theorem applies to this problem. What role does the strict increase of $g(t) = \lambda t^2$ play?

(b) Write the finite-dimensional optimization problem that results from substituting $f(x) = \sum_i \alpha_i k(x_i, x)$.

(c) Is this problem a quadratic program, a linear program, or something else? Explain.

Exercise 7.4 (Mercer Characterization Details)**.** ($**$)
Let k be a continuous pd kernel on $[0, 1]$ with Mercer expansion $k(x, x') = \sum_{j=1}^{\infty} \lambda_j \phi_j(x) \phi_j(x')$, and suppose $\lambda_j = j^{-3}$.

(a) Show that $f(x) = \sum_{j=1}^{\infty} j^{-3/2}\phi_j(x)$ does *not* belong to $\mathcal{H}_k$.

(b) Find the largest $\beta > 0$ such that $f(x) = \sum_j j^{-\beta}\phi_j(x)$ belongs to $\mathcal{H}_k$.

(c) Compute $\|f\|_{\mathcal{H}_k}$ for $f(x) = \sum_j j^{-2}\phi_j(x)$ and verify it is finite.

Exercise 7.5 (KRR Derivation with General Kernel)**.** $(**)$
Derive the KRR solution for the weighted loss $\sum_{i=1}^{n} w_i(y_i - f(x_i))^2 + \lambda\|f\|_{\mathcal{H}_k}^2$, where $w_i > 0$ are given weights. Show that the optimal $\alpha = (KWK+n\lambda K)^{-1}KWy$ where $W = \mathrm{diag}(w_1, \ldots, w_n)$, and simplify to a form involving only K and W. How does this reduce to standard KRR when $w_i = 1/n$ for all i?

Exercise 7.6 (Kernel Bellman Equation)**.** $(***)$
In a discounted MDP with finite state observations $s_1, \ldots, s_n$ and transitions (s_i, r_i, s_i'), the kernel Bellman equation seeks $\hat{V} \in \mathcal{H}_k$ minimizing

$$\sum_{i=1}^{n} \left(\hat{V}(s_i) - r_i - \gamma\hat{V}(s_i')\right)^2 + \lambda\|\hat{V}\|_{\mathcal{H}_k}^2.$$

(a) Apply the representer theorem to show that $\hat{V}(s) = \sum_j \alpha_j k(s_j, s)$.

(b) Derive the linear system that α must satisfy. Express it in terms of the Gram matrices $K_{ij} = k(s_i, s_j)$ and $K'_{ij} = k(s_i, s_j')$.

(c) Under what conditions on γ, λ, and k is this system guaranteed to have a unique solution?

Hint: The representer theorem applies since the objective depends on $\hat{V}$ only through evaluations at $s_1, \ldots, s_n$ and $s_1', \ldots, s_n'$. Augment the set of kernel sections.

8 Summary

- A **reproducing kernel Hilbert space** (RKHS) is a Hilbert space of functions where point evaluation is a bounded linear functional. The RKHS norm controls pointwise values: $|f(x)| \leq \|f\|_{\mathcal{H}_k}\sqrt{k(x,x)}$.
- Every pd kernel k determines a **unique** RKHS $\mathcal{H}_k$ (**Aronszajn's theorem**), constructed by completing the span of kernel sections $\{k(x, \cdot)\}$ with the inner product $\langle k_x, k_{x'}\rangle = k(x, x')$.
- The **reproducing property** $f(x) = \langle f, k_x\rangle_{\mathcal{H}_k}$ allows pointwise evaluation via inner products, connecting function evaluation to Hilbert space geometry.
- Under the Mercer expansion, $f \in \mathcal{H}_k$ if and only if $\sum_j a_j^2/\lambda_j < \infty$, where a_j are the expansion coefficients. The RKHS norm $\|f\|_{\mathcal{H}_k}^2 = \sum_j a_j^2/\lambda_j$ penalizes high-frequency components, imposing a **smoothness bias**.
- The **representer theorem** guarantees that the minimizer of any regularized empirical risk over $\mathcal{H}_k$ lies in $\mathrm{span}\{k(x_1, \cdot), \ldots, k(x_n, \cdot)\}$, reducing infinite-dimensional optimization to an n-dimensional problem.
- **Kernel ridge regression** has the closed-form solution $\alpha^* = (K + n\lambda I)^{-1}y$, with excess risk $O(n^{-1/2})$ in the worst case and faster rates when the kernel eigenvalues decay rapidly.
- The optimal regularization parameter $\lambda^* = O(n^{-1/2})$ balances approximation error ($O(\lambda B^2)$, decreasing in λ) against estimation error ($O(\kappa^2 M^2/(n\lambda))$, increasing in λ).
- KRR with a Gaussian kernel achieves near-parametric rates $O(n^{-1}\log n)$ due to exponential eigenvalue decay, while the Sobolev kernel achieves $O(n^{-2/3})$ due to polynomial decay ($\lambda_j = O(j^{-2})$).

9 Further Reading

- **Lesson 63 (RNN)** develops recurrent neural networks, which operate on sequential data. The kernel perspective on neural networks (via the NTK) connects the RKHS framework to the analysis of training dynamics in deep learning.
- Steinwart and Christmann [2008] provide a comprehensive and rigorous treatment of RKHS theory, including universal kernels, approximation properties, and oracle inequalities for SVMs.
- Bach [2024] develops kernel ridge regression rates with eigenvalue-dependent bounds, including minimax-optimal results and comparisons with neural network methods.
- Schölkopf and Smola [2002] cover the practical aspects of kernel methods, including kernel selection, multi-task learning, and kernel methods for structured data.
- Jacot et al. [2018] and Lee et al. [2019] establish the connection between wide neural networks and kernel methods via the neural tangent kernel, showing that infinite-width networks perform kernel ridge regression in the NTK RKHS.
- The RKHS theory was introduced by Aronszajn [1950] in the context of the theory of reproducing kernels. The representer theorem in the generality presented here is due to Schölkopf, Herbrich, and Smola (2001), building on earlier work by Kimeldorf and Wahba (1970) for spline smoothing.

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